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ROLLNO.: 20

COURSE: AI&DS

CLASS: BE

SUB: Computer Laboratory-III

Title: Implementation of Fuzzy Set Operations and Fuzzy Relations.

Code:

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# -----  
# Fuzzy Set Operations Functions  
# -----  
def fuzzy_union(A, B):  
    return {x: max(A[x], B[x]) for x in A}  
def fuzzy_intersection(A, B):  
    return {x: min(A[x], B[x]) for x in A}  
def fuzzy_complement(A):  
    return {x: 1 - A[x] for x in A}  
def fuzzy_difference(A, B):  
    return {x: min(A[x], 1 - B[x]) for x in A}  
# -----  
# Fuzzy Relation (Cartesian Product)  
# -----  
def fuzzy_cartesian_product(A, B):  
    R = {}  
    for a in A:  
        for b in B:  
            R[(a, b)] = min(A[a], B[b])  
    return R  
# -----  
# Max-Min Composition of Relations  
# -----
```

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def max_min_composition(R, S, A_elements, B_elements, C_elements):
    T = {}
    for a in A_elements:
        for c in C_elements:
            values = []
            for b in B_elements:
                values.append(min(R[(a, b)], S[(b, c)]))
            T[(a, c)] = max(values)
    return T

# -----
# Main Program
# -----

# Define fuzzy sets
A = {'x1': 0.2, 'x2': 0.6, 'x3': 0.8}
B = {'x1': 0.5, 'x2': 0.4, 'x3': 0.9}
print("Fuzzy Set A:", A)
print("Fuzzy Set B:", B)
print("\nUnion:", fuzzy_union(A, B))
print("Intersection:", fuzzy_intersection(A, B))
print("Complement of A:", fuzzy_complement(A))
print("Difference (A - B):", fuzzy_difference(A, B))

# Define fuzzy sets for relations
A_rel = {'a1': 0.6, 'a2': 0.8}
B_rel = {'b1': 0.5, 'b2': 0.7}
C_rel = {'c1': 0.4, 'c2': 0.9}

# Create fuzzy relations
R = fuzzy_cartesian_product(A_rel, B_rel)
S = fuzzy_cartesian_product(B_rel, C_rel)
print("\nFuzzy Relation R ( $A \times B$ ):")
for k, v in R.items():
    print(k, ":", v)

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print("\nFuzzy Relation S ( $B \times C$ ):")
for k, v in S.items():
    print(k, ":", v)
# Max–Min Composition
T = max_min_composition(
    R, S,
    A_elements=['a1', 'a2'],
    B_elements=['b1', 'b2'],
    C_elements=['c1', 'c2']
)
print("\nMax–Min Composition (T):")
for k, v in T.items():
    print(k, ":", v)

```

Output:

```

Fuzzy Set A: {'x1': 0.2, 'x2': 0.6, 'x3': 0.8}
Fuzzy Set B: {'x1': 0.5, 'x2': 0.4, 'x3': 0.9}
Union: {'x1': 0.5, 'x2': 0.6, 'x3': 0.9}
Intersection: {'x1': 0.2, 'x2': 0.4, 'x3': 0.8}
Complement of A: {'x1': 0.8, 'x2': 0.4, 'x3': 0.19999999999999996}
Difference (A - B): {'x1': 0.2, 'x2': 0.6, 'x3': 0.09999999999999998}
Fuzzy Relation R ( $A \times B$ ):
('a1', 'b1') : 0.5
('a1', 'b2') : 0.6
('a2', 'b1') : 0.5
('a2', 'b2') : 0.7
Fuzzy Relation S ( $B \times C$ ):
('b1', 'c1') : 0.4
('b1', 'c2') : 0.5
('b2', 'c1') : 0.4

```

$('b2', 'c2') : 0.7$

Max–Min Composition (T):

$('a1', 'c1') : 0.4$

$('a1', 'c2') : 0.6$

$('a2', 'c1') : 0.4$

$('a2', 'c2') : 0.7$